

# Peh Hoe Khim Marcus

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Performance-driven Software Engineer with experience in designing, developing, and deploying robust software solutions. Skilled in problem-solving and time management, with a strong focus on delivering high-quality results.

## WORK EXPERIENCES

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| <b>TikTok, Backend Software Engineer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Apr 2024 - Present</b>                       |
| <ul style="list-style-type: none"><li>Designed and launched widget table with visualization query support, enabling low-code solution for non-technical users; scaled to cover 70%+ of table use cases across TikTok Live</li><li>Led development of a global user analysis feature from scratch, including cross-region data pipelines, backend logic, and orchestration for dynamic cohort refinement</li><li>Implemented Redis caching for data widgets, enhancing request response times by 30% and improving system scalability under high-load conditions</li></ul> |                                                 |
| <b>ByteDance, Backend Software Engineering Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Aug 2023 – Dec 2023</b>                      |
| <ul style="list-style-type: none"><li>Overhauled approval process by integrating in-house solution, increasing efficiency &gt;30%</li><li>Contributed to channel platform with monthly transactions exceeding USD \$7.5 billion</li><li>Raised performance of channels rating API by over 90% through effective optimization strategies</li></ul>                                                                                                                                                                                                                         |                                                 |
| <b>Google, Software Engineering Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>May 2023 - Aug 2023</b>                      |
| <ul style="list-style-type: none"><li>Refined onboarding verification flow on Google Pay’s merchant application (Android)</li><li>Optimized QR code loading time by 93.5% (from 1.38s to 0.09s) through elimination of RPC calls</li><li>Lowered proximity verification range by 99.5% from 2km to 10m by leveraging Bluetooth Low Energy</li></ul>                                                                                                                                                                                                                       |                                                 |
| <b>GovTech, Software Engineering Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>May 2022 - Jul 2022, Dec 2022 - Jan 2023</b> |
| <ul style="list-style-type: none"><li>Reduced loading time of web page by 10% by removing unnecessary API calls</li><li>Strengthened security by implementing Two-Factor Authentication</li><li>Improved server query response time under load by 200x through API restructuring</li></ul>                                                                                                                                                                                                                                                                                |                                                 |

## EDUCATION

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|-----------------------------------------------------------|-----------------------------------------------|
| <b>Master of Science in Computer Science</b>              | <b>Aug 2024 – Present</b>                     |
| Georgia Institute of Technology                           | <b>GPA: 4.00 / 4.00</b>                       |
| <b>Bachelor of Computing in Computer Science (Honors)</b> | <b>Aug 2020 – Jan 2024</b>                    |
| National University of Singapore                          | <b>GPA: 4.78 / 5.00 (Highest Distinction)</b> |

## PROJECTS

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| <b>Instrument Tracking</b> – GIC Code to Impact 2022   2 <sup>nd</sup> runner up                                                                                                                                                                                                                                                            |
| <ul style="list-style-type: none"><li>Spearheaded team of 5 to delivered an instrument tracking web application catered to business</li><li>Added extra features such as cross checking of data and logging system</li><li>Constructed using <i>Node</i>, <i>NextJS</i> and <i>Postgres</i></li></ul>                                       |
| <b>Around the World in \$80</b> - Expense splitting    Project                                                                                                                                                                                           |
| <ul style="list-style-type: none"><li>Targets pain point of group travelers in splitting bills</li><li>Use various software design pattern such as "command pattern" and "model, view, controller"</li><li>Programmed in <i>Java</i> with <i>JavaFx</i> Library with 90% code coverage</li></ul>                                            |
| <b>Inclusive Search</b> – Google HelloAI (internal) hackathon   Finalist                                                                                                                                                                                                                                                                    |
| <ul style="list-style-type: none"><li>Built LLM prototype for more inclusive search results, incorporating diverse cultural perspectives</li><li>Pitched the AI solution, showcasing strong communication and presentation skills</li><li>Ranked among the top 10 teams, demonstrating teamwork and excellence in problem-solving</li></ul> |

## SKILLS & CERTIFICATIONS

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**Programming Languages:** Python, C++, Go, JavaScript, TypeScript, Java, SQL  
**Technologies:** ReactJs, NextJs, Flutter, React Native, Postgres, MySQL, MongoDB  
**Certifications:** AWS Solution Architecture Associate